

## Copilot Studio Training – Day 10

### Agenda:

- Complex Autonomous Agent with Auto Reply from Excel Sources
- Dynamic Chaining in Agent
- Connect with Agent Flows
- Custom Connector for Autonomous Agent
- Create Custom Tool for Copilot Agent

### Use Case 01 – Autonomous Agent with Excel Files and Custom Connector

#### Objective

The goal of this exercise is to understand and implement **power of Knowledge Sources like Excel** and **Custom Connector** in Copilot Studio. Suppliers will send quotation requests via email, typically including multiple product names or product codes in their messages. Your system must ensure that no requested product is missed. Once they get the price they need to perform some mathematical calculations and need to convert the price from USD to INR using the Custom Connector.

#### Requirements

Develop an automated process that performs the following steps when a new email with the subject "**Request for a Price**" is received:

1. **Extract Product Details:**
  - Parse the email body to identify **all** product codes or product names mentioned.
2. **Retrieve Product Information:**
  - Access the internal knowledge base (KB) to retrieve the following details for each product:
    - **Product Name**
    - **Product ID**
    - **Price in USD**
    - **Stock Status**
3. **Currency Conversion:**
  - Use the **Get Currency Exchange Rate** action to fetch the latest exchange rate from **USD to INR**.
  - Convert the price from USD to INR using this exchange rate.

#### 4. Generate an Email Response:

- Format the response as an **HTML table** containing:
  - **Product ID**
  - **Product Name**
  - **Price in USD**
  - **Converted Price in INR**
  - **Stock Status**
- Include a **customer support message**:  
*"For further inquiries, feel free to call our customer care department at **555 1234**."*
- Sign off the email with:  
*"Best regards,  
**Price Mate**"*

#### 5. Send the Email:

- Ensure proper **HTML formatting** to make the email visually structured and readable.
- Send the formatted email to the user.

Implement Custom Connector: Get Currency Exchange Rate  
<https://currencylayer.com/>

#### Solution

Agent Name: PriceMate Request for Price

Description:

You're Price mate. You're task is to help the suppliers or customers to provide the quotation for the products they're looking for.

Instructions:

You are created to help to suppliers for helping them with the price quotation. They are sending the request of quotation via email. Mostly they're having multiple products with their product number or product names. Make sure you'll not miss any product from the email. You should follow the steps below without missing any. Once all actions are completed your work is finished. When you receive a new email with the subject line "Request for a Price" perform the following tasks without asking the user for information.

1. Parse the body of the email to identify the all Product codes or Product name they've asked for the quote.
2. Use the internal KB Parts Price Information to find out the Price details. From internal KB get Product Name, Product ID, Price and Stock Status.
3. Use action Currency Conversion to find out currency rate in the appropriate currency user has asked for in the email.
4. Convert the USD to appropriate currency conversion rate.

5. Write an email with the proper structure with Product ID, Product Name, Price in USD, Price in appropriate currency user has asked and Stock Status. Use Proper HTML table structure to send an email using Send an email for Price Quotation . Please also suggest they can call our customer care department on 555 1234. Use a friendly sign off as "Price Mate". Please do proper formatting of the email with html and send formatted email to user.

Email for Testing:

Hey Price Mate,

Hello, I need the quotation in INR, GBP and EURO for below parts.

CAR-PART-0020  
CAR-PART-0021  
CAR-PART-0022  
CAR-PART-0023  
CAR-PART-0040  
CAR-PART-0041  
CAR-PART-0042

Output:

Price Quotation for Requested Car Parts 👉 Inbox x

 **Dhruvin Shah**  
to me ▾

Hey Dhruvin, 

Thank you for reaching out. Here are the price quotations for the requested car parts:

| Product ID    | Product Name              | Price (USD) | Price (INR) | Stock Status |
|---------------|---------------------------|-------------|-------------|--------------|
| CAR-PART-0020 | Side Mirror - Model 20    | 424.55      | 36800.92    | In Stock     |
| CAR-PART-0021 | Steering Wheel - Model 21 | 649.36      | 56312.92    | In Stock     |
| CAR-PART-0022 | Alternator - Model 22     | 537.31      | 46575.92    | In Stock     |
| CAR-PART-0023 | Carburetor - Model 23     | 418.67      | 36288.92    | In Stock     |
| CAR-PART-0040 | Wheel Bearings - Model 40 | 30.78       | 2667.92     | In Stock     |
| CAR-PART-0041 | Timing Belt - Model 41    | 918.61      | 79657.92    | In Stock     |
| CAR-PART-0042 | Radiator - Model 42       | 970.05      | 83947.92    | In Stock     |

If you have any further questions, feel free to call our customer care department at 555 1234.

Best regards,  
Price Mate

## Key Takeaways

By completing this exercise, you will learn:

- ✓ A clear understanding of **Excel as Knowledge Source** and their role in Copilot workflows.
- ✓ How to **implement Custom Connector?**.
- ✓ The process of combining the result and sending final responses.

## Use Case 02: Understanding Dynamic Chaining with Auto PO Creation Autonomous Agent [SharePoint + Agent Flows + Custom Connector]

### Objective:

The goal is to automate the **Purchase Order (PO) creation and approval process** efficiently, ensuring that no requests are missed. The system should extract relevant details from user requests, create a PO record, initiate an approval workflow, perform currency conversion, and send a professional summary email.

### Requirements:

This system enables users to submit multiple purchase order requests, which are processed sequentially. The workflow consists of the following steps:

#### 1. Extract PO Details:

- Read the received message to capture key information, including:
  - Requestor's Name (Logged-in User)
  - Supplier Name
  - Unit Price (USD)
  - Quantity Ordered

#### 2. Create PO in SharePoint:

- Call the **Create Item** action to store the PO record in SharePoint.

#### 3. Initiate PO Approval Process:

- Calculate the **Total PO Amount** using:  
**Total Amount = Unit Price × Quantity Ordered**
- Trigger the **PO Approval action**, which includes:
  - Sending an approval request to relevant stakeholders.
  - Providing an **HTML Summary** containing:
    - PO ID
    - Supplier Name
    - Quantity Ordered
    - Unit Price (USD)
- If overall amount is > 10000 then it will send an approval email and status will be "Pending". If not, it simply maintains a status "Auto Approved"
- The approver can either **approve or reject** the PO, and their decision will determine the next steps.

#### 4. Currency Conversion:

- Upon PO creation, retrieve the **current exchange rate** using the **GetCRate** custom connector.
- Convert the PO Amount from **USD to INR** for accurate financial representation.

#### 5. **Send Summary Email:**

- Generate a **professional HTML email** containing a well-structured table with the following details:
  - Supplier Information
  - Quantity Ordered
  - Unit Price (USD)
  - Converted Unit Price (INR)
  - PO Approval Status
  - PO ID
- Use the **Send Email action** to notify the current user with the summary.

#### Key Learning Points:

- Trigger and Actions
- Knowledgebase as Public Facing website
- SharePoint connections

#### Solution Prompt:

**Agent Name:** PO Creation DC

#### Description:

You're helping us to create a Purchase Order based on the information supplied.

#### Instruction:

Your task is to create a Purchase order into the system. Users can send you multiple purchase order request. Make sure you do not miss any Purchase order. You need to follow below tasks in the sequence.

1. Read the information from the received message and try to find out the purchase order information like, a person who requested PO, supplier name, unit price and qty. Here Title is current logged in User's Name.
2. Call an action Create item to create the record into SharePoint.
3. With this start PO Approval process by calling PO Approval action. PO Amount will be Unit Price multiply by Qty Ordered. Send overall Summary generated in proper HTML format including PO ID, Supplier Name, QTY and Unit Price in Approval action.
4. Send Summary email in table format with Supplier Information, Qty Ordered, Unit Price \$, Unit Price INR, PO Approval Status and ID. Use proper HTML table format with professional look and send to current user using Send Email action.

Test Messages:

**Sample 1:**

I want to create 2 po with below information:

Supplier: Neela Telefilms  
Qty: 25 Price: 500

Supplier: TRS Productions  
Qty: 50 Price: 750

**Sample 2:**

I want to create 2 po with below information:

Supplier: Sagarika Enterprise  
Qty: 10 Price: 50

Supplier: Mangesh Corp  
Qty: 50 Price: 7.5

Output:

Purchase Order Summary

DS

Dhruvin Shah

To: Dhruvin Shah

Sat 2/1/2025 8:00 PM

Purchase Order Summary

| PO ID | Supplier  | Qty Ordered | Unit Price \$ | Unit Price INR | PO Approval Status |
|-------|-----------|-------------|---------------|----------------|--------------------|
| 22    | ABC Crp   | 25          | 500           | 43348.675      | Pending Approval   |
| 23    | Delta Crp | 50          | 750           | 65022.75       | Pending Approval   |

Key Learning Points:

- Agent Flow Integration
- Custom Connector Implementation
- Dynamic Chaining Concept

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